9.2 DISADVANTAGES :

1. Dependence on Dataset Quality and Diversity:

The accuracy and reliability of the HematoVision model are directly dependent on the quality, size, and diversity of the training dataset. If the dataset contains blurry images, improper labeling, class imbalance, or limited variations in cell types, the model may not learn representative features effectively. This can result in biased predictions or reduced accuracy when encountering new data. Therefore, continuous data collection and dataset improvement are essential.

2. Limited Generalization Across Different Environments:

The model may perform well on images captured under specific laboratory conditions but struggle when exposed to images from different microscopes, lighting setups, or staining techniques. Variations in image resolution, background noise, and contrast can affect prediction accuracy. Without proper domain adaptation or retraining, the model may not generalize effectively across multiple healthcare settings.

3. Computational Resource Requirements:

Although transfer learning reduces overall training time, deep learning models still require substantial computational resources. Training the model efficiently often requires a GPU with CUDA support. In resource-limited laboratories or rural healthcare centers, access to high-performance hardware may be limited. This can restrict real-time deployment or slow down system performance.

4. Risk of Misclassification and False Predictions:

No machine learning model guarantees 100% accuracy. There is always a risk of false positives or false negatives. For example, similar-looking blood cells may be incorrectly classified, especially in borderline or abnormal cases. In medical diagnosis, even small misclassifications can have serious consequences. Therefore, the system must always be used as a supportive tool rather than a standalone diagnostic solution.

5. Lack of Interpretability (Black-Box Nature):

Deep learning models such as MobileNetV2 operate as complex neural networks with millions of parameters. It can be difficult to clearly explain how the model arrives at a particular prediction. This lack of transparency may reduce trust among healthcare professionals who prefer explainable and interpretable diagnostic tools. Advanced explainability techniques may be required to address this concern.

6. Continuous Maintenance and Model Updates:

Medical data and diagnostic standards evolve over time. The HematoVision model may require periodic retraining with updated datasets to maintain accuracy and relevance. Without proper maintenance, model performance may degrade. Continuous monitoring, validation, and updating are necessary for long-term reliability.

7. Dependence on Technical Infrastructure:

If deployed as a web application, the system depends on stable server infrastructure and internet connectivity. Technical failures, server downtime, or software bugs can temporarily disrupt access to predictions. This may affect laboratory workflows if alternative manual methods are not available.

8. Ethical and Legal Considerations:

Using AI in healthcare raises ethical and legal concerns, including data privacy, patient confidentiality, and accountability for incorrect predictions. Proper data protection mechanisms and regulatory compliance must be ensured to prevent misuse or data breaches.

9. Not a Substitute for Professional Expertise:

HematoVision is designed to assist healthcare professionals, not replace them. Final diagnosis and treatment decisions must always be made by qualified medical experts. Over-reliance on automated systems without professional review could pose risks to patient safety.

Although HematoVision provides automation, speed, and improved efficiency in blood cell classification, it is important to recognize its limitations. Addressing these disadvantages through continuous improvement, dataset expansion, proper validation, and professional supervision will ensure safer and more effective implementation in real-world healthcare environments.